2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



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BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

Engr. Shanzay Jamil

Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING FACULTY OF ENGINEERING AIR UNIVERSITY, ISLAMABAD

	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 13

Lab Title: IMPLEMENTATION AND VERIFICATION OF THEVENIN THEOREM BOTH ON HARDWARE AND SOFTWARE

Objectiv: THEVENIN THEOREM

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 8/May/2025 Signature:

LABORATORY
EXPERIMENT
NO. 13

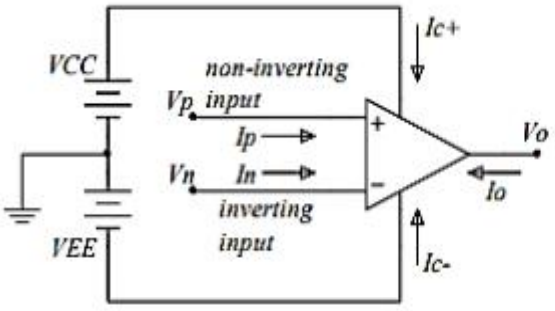
Inverting and on-inverting OP-AMP

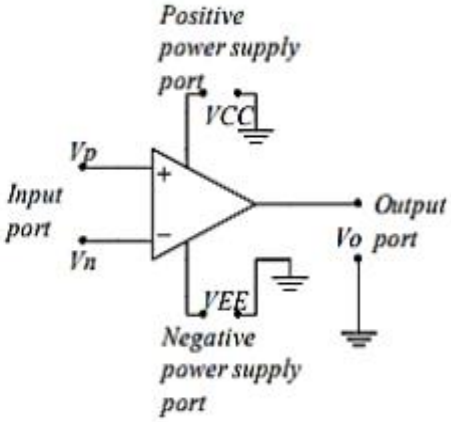
OBJECTIVE:

- To implement op-amp as an inverting amplifier.
- To implement op-amp as a non-inverting amplifier.
- To verify the results through software using Proteus.

Operational Amplifier

The operational amplifier (op-amp) is a voltage controlled voltage source with very high gain. It is a five terminal four port active element. The symbol of the op-amp with the associated terminals and ports is shown on Figure 1(a) and (b).

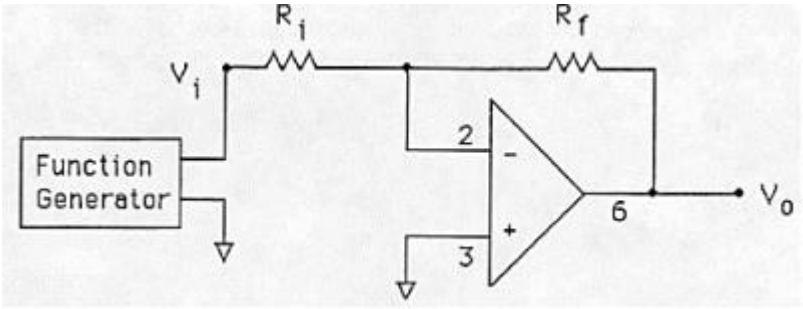




Lab Tasks:

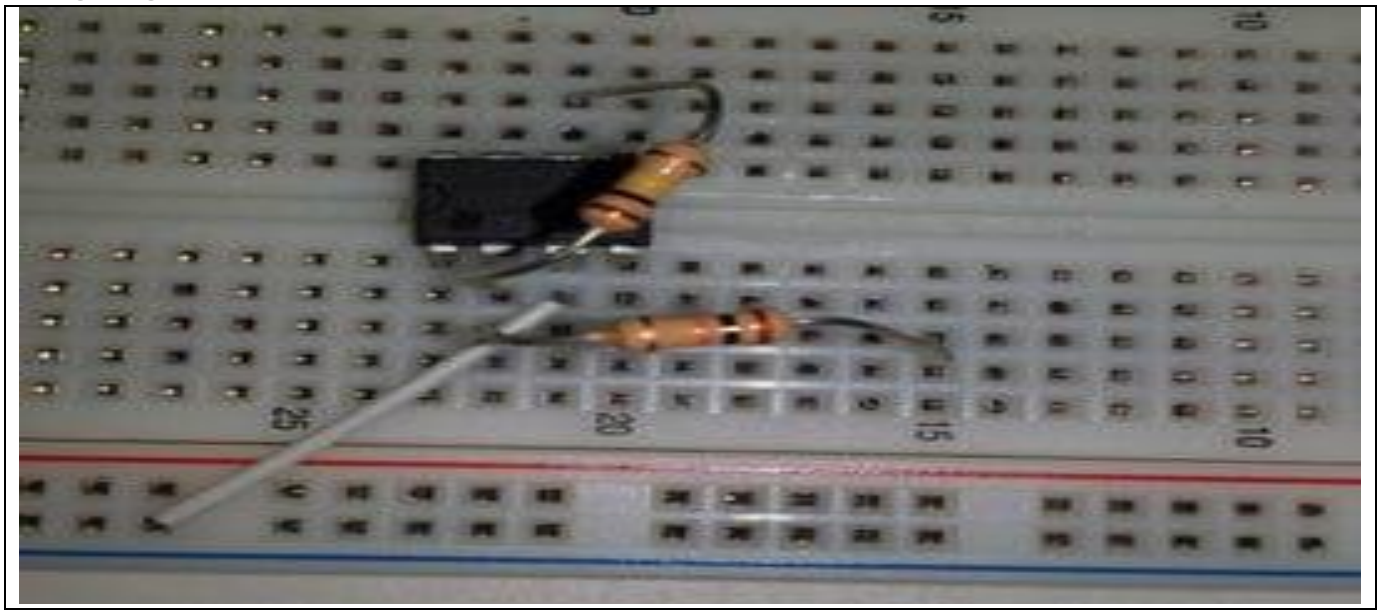
1. Inverting Amplifier:

Construct and design an inverting amplifier with three gains (-1, -10, -21).



Start with choosing Ri and Rf. Use the function generator LOW OUTPUT to supply a 1 kHz 0.1V peak-to-peak sine wave to the input (Vi) to pin 3. Use oscilloscope CH1 to measure Vi and CH2 to measure Vo. You may find it useful to trigger the oscilloscope sweep using the function generator SYNC OUTPUT as an external trigger.

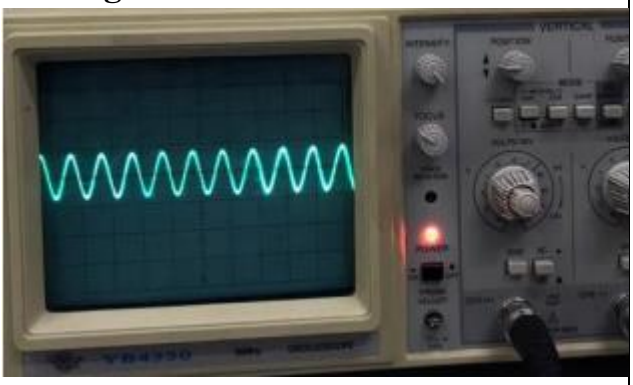
Designing:



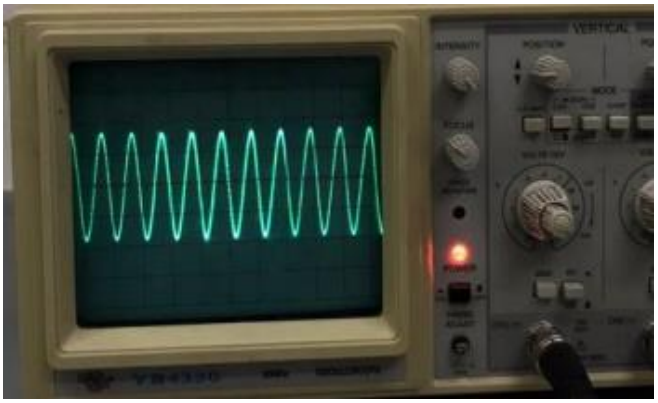
Experimental Results:

INPUT VOLTAGE:	Gain	OUTPUT VOLTAGE
1V	6V	6V
2V	11V	12V
3V	19V	18V
4V	20V	24V
5V	21V	30V

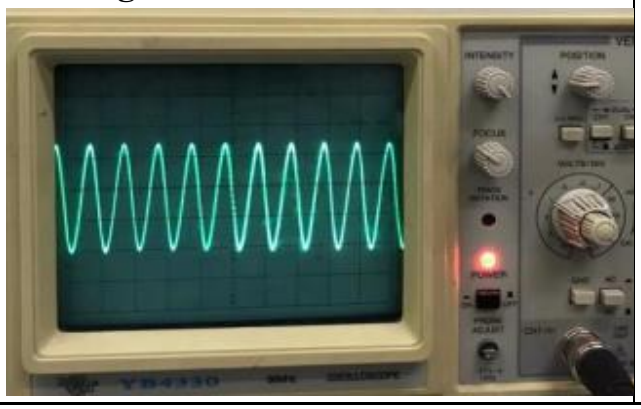
1Volatge:



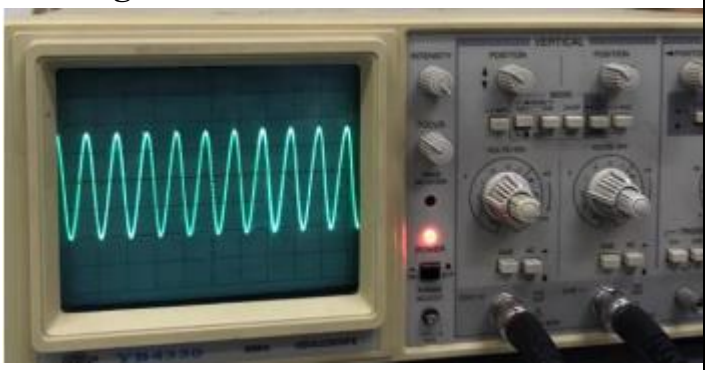
2 Voltage

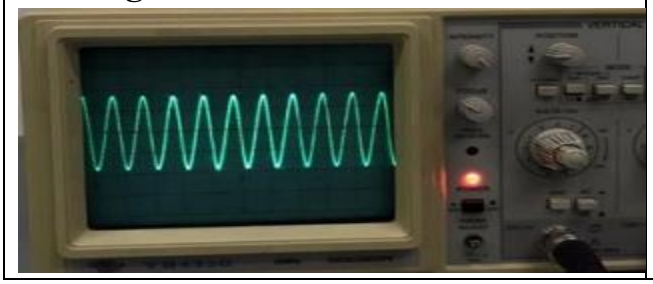


3 Voltage



4 Voltage



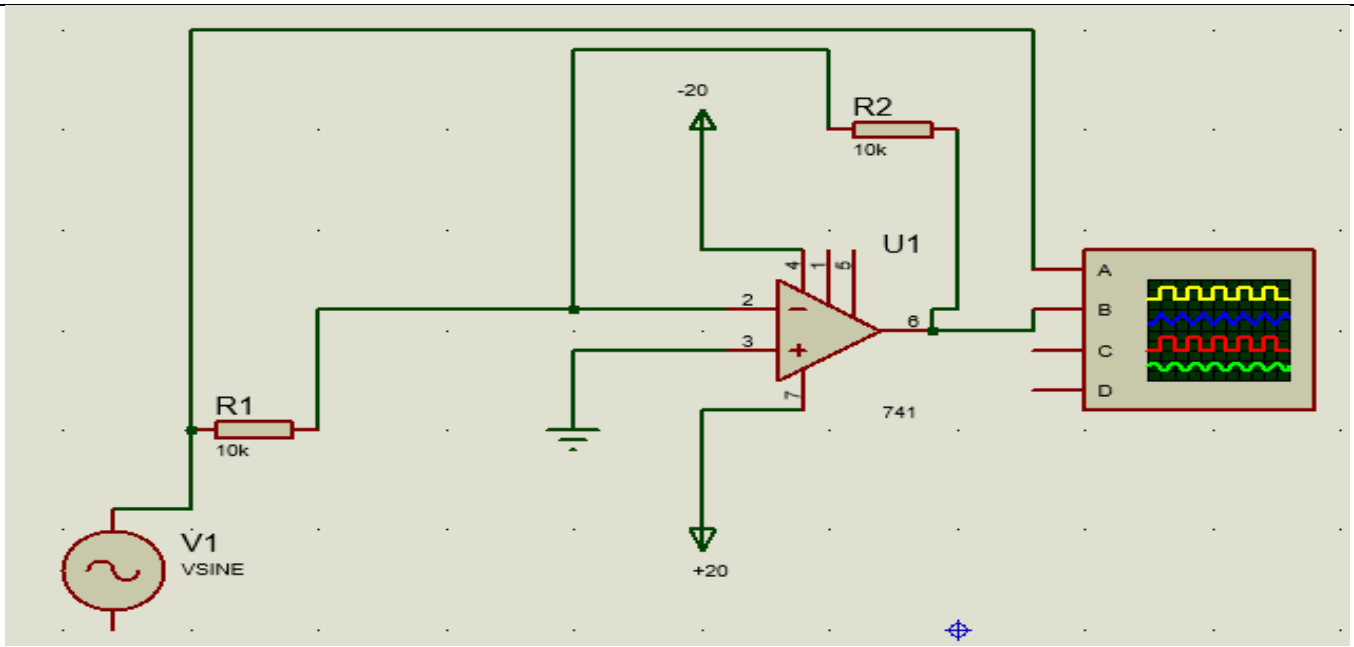
5 voltage: 	**In inverting amplifier there is shift of 180 degree in sin-waves and waves shifdt in 180degree and also increases voltage and give the maximum gain** **Gain=Vout/Vin**

Theoretical Calculation:

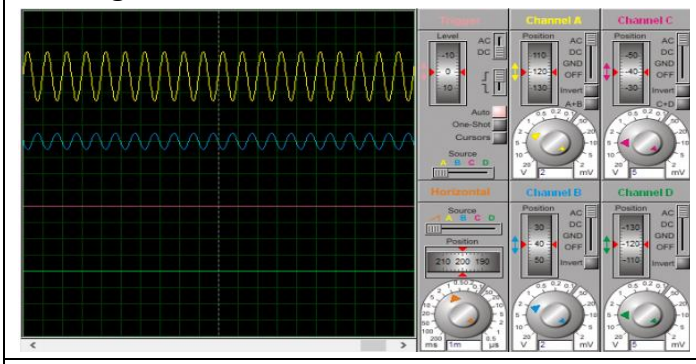
OUTPUT VOLTAGE	**GAIN**
Given the non-inverting amplifier formula: Vout=Vin×(1+RinRf) And the assumed gain: RinRf=5⟹Gain=(1+RinRf)=(1+5)=6 Now, let's calculate Vout for each given Vin: 1. For Vin=1V: Vout=1V×6=6V 2. For Vin=2V: Vout=2V×6=12V 3. For Vin=3V: Vout=3V×6=18V 4. For Vin=4V: Vout=4V×6=24V 5. For Vin=5V:	Given the formulas: Gain=VinVout Vout=Vin×Gain Let's calculate for each pair of Vin and Vout values: 1. For Vin=1V, Vout=6V Gain=1V6V=6 Vout=1V×6=6V 2. For Vin=2V, Vout=11V Gain=2V11V=5.5 Vout=2V×5.5=11V 3. For Vin=3V, Vout=19V Gain=3V19V≈6.33 Vout=3V×6.33=18.99V 4. For Vin=4V, Vout=20V Gain=4V20V=5 Vout=4V×5=20V 5. For Vin=5V, Vout=21V

Vout=5V×6=30V	Gain=5V21V=4.2 Vout=5V×4.2=21V

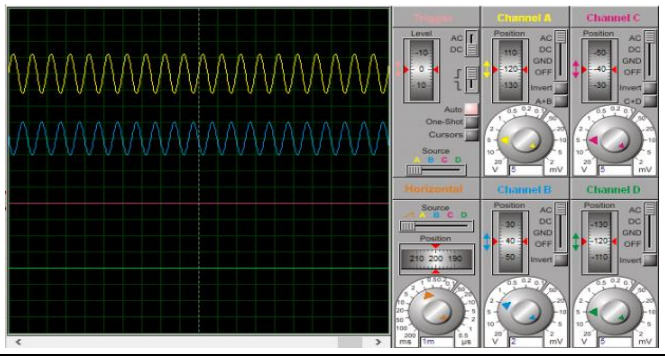
Protius Simulation:



1 Voltage;

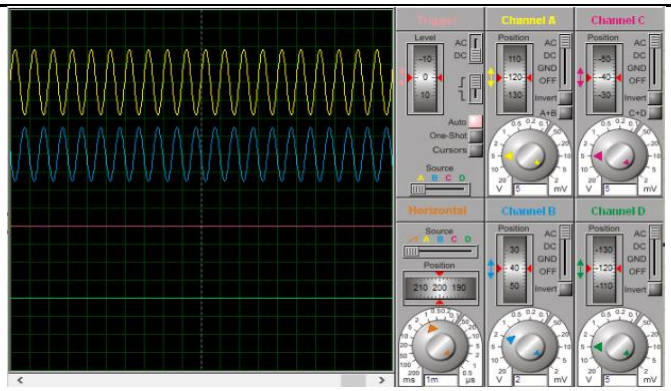


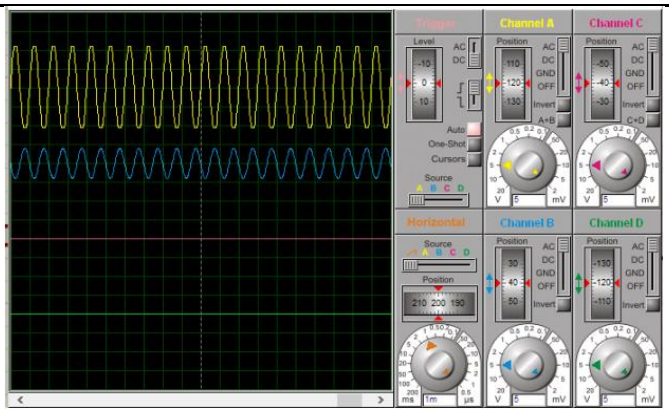
2 Voltage:



3 Voltage:

4 Voltage:

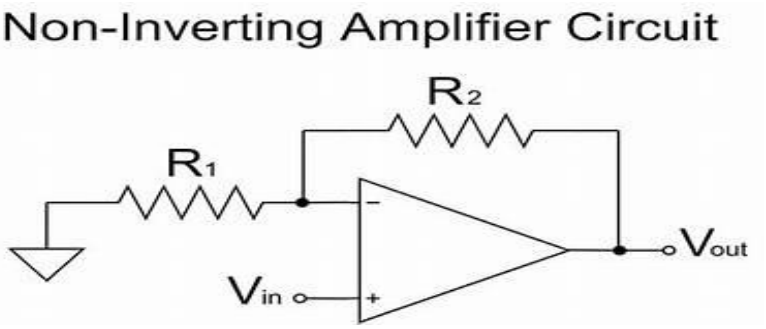




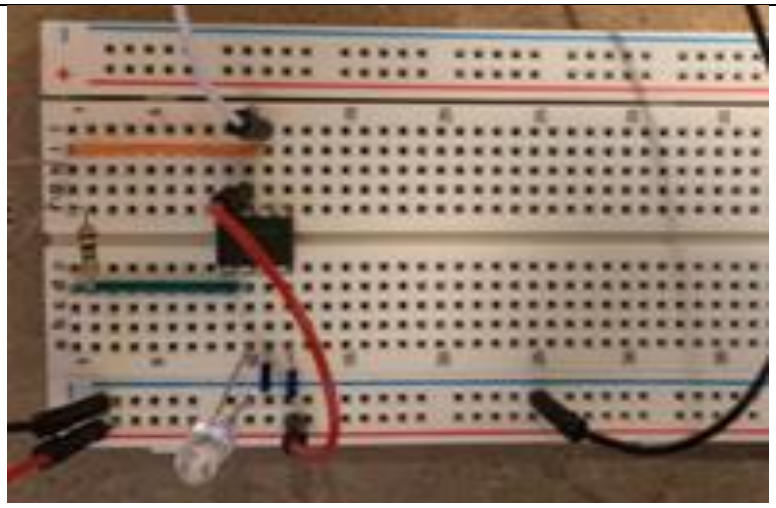
Task 2:

2. Non-Inverting Amplifier:

Construct and design an inverting amplifier with three gains (1, 10, 13).



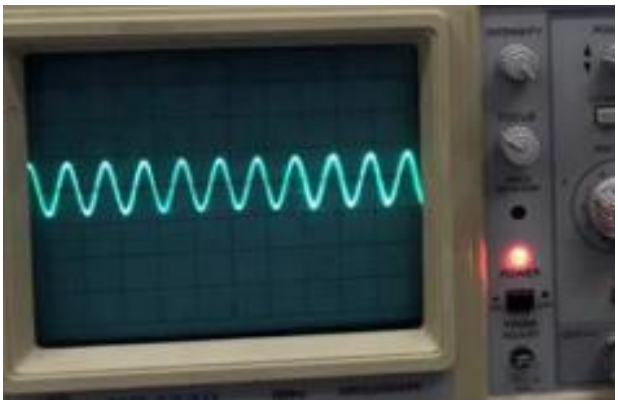
Design:



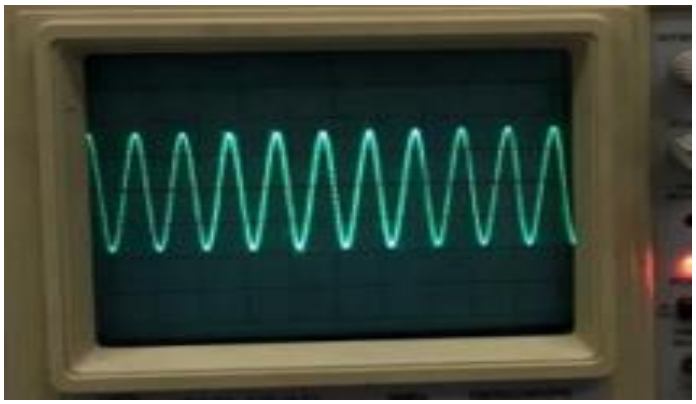
Experimental Results:

INPUT VOLTAGE:	Gain	OUTPUT VOLTAGE
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3V	19V	18V
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5V	21V	30V

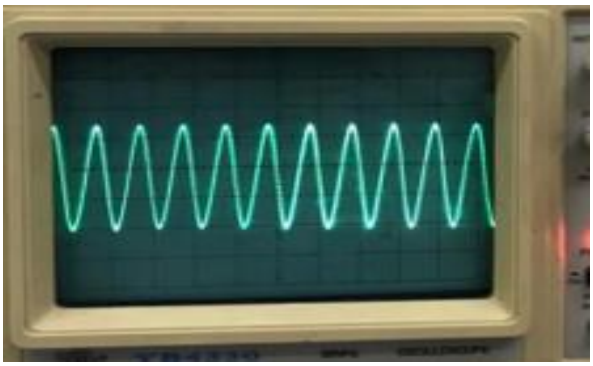
1Volatge:



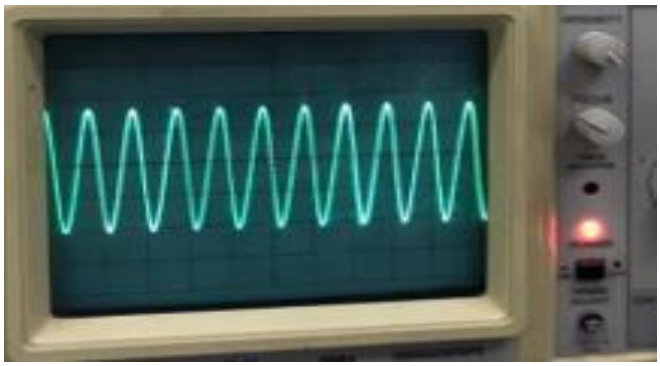
2 Voltage



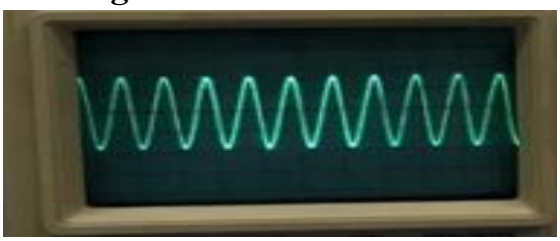
3 Voltage



4 Voltage



5 voltage:

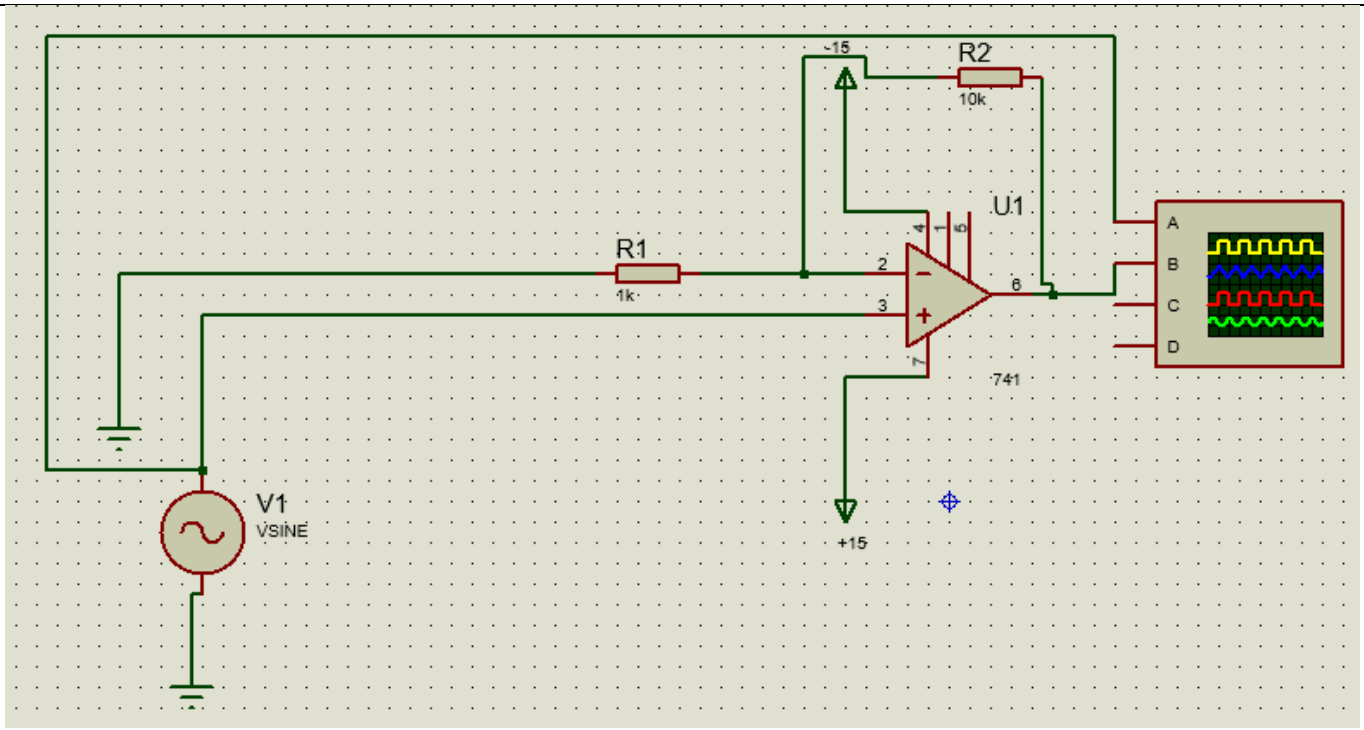


In non_inverting sin waves remain same but amplitude and frequency changes. And voltages output increases anmd gain maximu.There is no shift of 180 degree

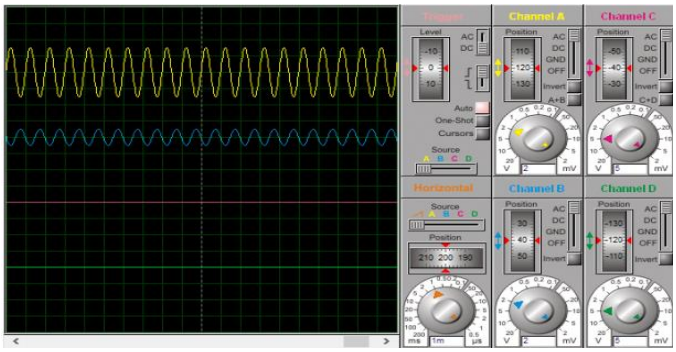
Theoretical Calculation:

OUTPUT VOLTAGE(Vo)	GAIN (G)
Given the formulas: **Av=VinVoutAv=1+RinRfVout=Vin·Av**	**Given the formulas:** **Gain=VinVout** **Vout=Vin×Gain**
1. Vin=1V, Vout=6V Av=1V6V=6 6=1+RinRf ⇒RinRf=5 Vout=1V·6=6V	1. For Vin=1V, Vout=6V Gain=1V6V=6 Vout =1V×6=6V
2. Vin=2V, Vout=11V Av=2V11V=5.5 5.5=1+RinRf⇒RinRf=4.5 Vout=2V·5.5=11V	2. For Vin=2V, Vout=11V Gain=2V11V=5.5 Vout=2V×5.5=11V
3. Vin=3V, Vout=19V Av=3V19V≈6.33 6.33=1+RinRf⇒RinRf≈5.33 Vout =3V·6.33≈18.99V	3. For Vin=3V, Vout=19V Gain=3V19V≈6.33 Vout=3V×6.33=18.99V
4. Vin=4V, Vout=20V Av=4V20V=5 5=1+RinRf ⇒RinRf=4 Vout=4V·5=20V	4. For Vin=4V, Vout=20V Gain=4V20V=5 Vout =4V×5=20V
5. Vin=5V, Vout=21V Av=5V21V=4.2 4.2=1+RinRf⇒RinRf=3.2 Vout=5V·4.2=21V	5. For Vin=5V, Vout=21V Gain=5V21V=4.2 Vout=5V×4.2=21V

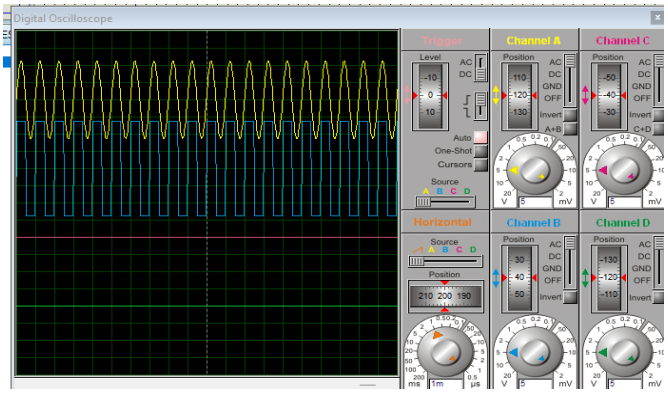
Protius Simulation:



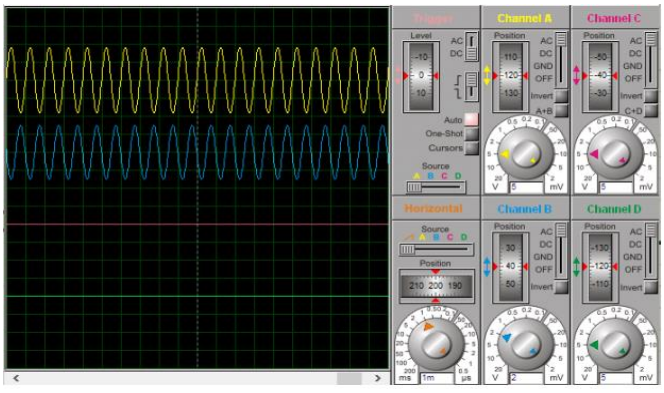
1 Voltage;



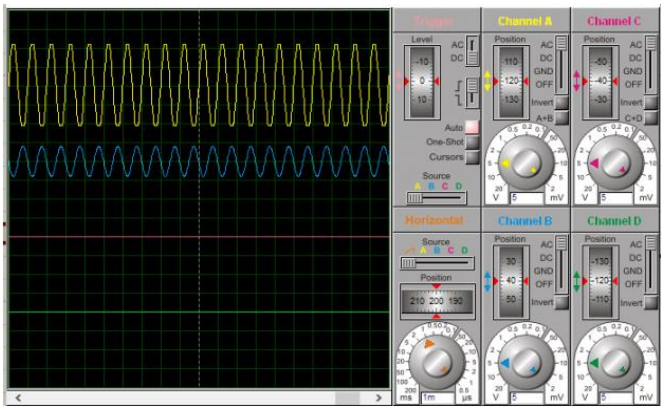
2 Voltage:



3 Voltage:



4 Voltage:



Conclusions:

In the realm of operational amplifiers, both the inverting and non-inverting configurations serve as essential tools for signal amplification, yet they differ fundamentally in their output phase relationship with the input. The **inverting op-amp** amplifies an input signal while simultaneously flipping its polarity, meaning a positive input results in a negative output and vice-versa, making it suitable for applications requiring signal inversion. Conversely, the **non-inverting op-amp** amplifies the input signal without altering its phase, delivering an output that retains the same polarity as the input. Therefore, the choice between these two ubiquitous configurations hinges entirely on whether the circuit design necessitates a phase reversal of the amplified signal or its preservation.